Car

- brand: string
- model: string
- year: int
- price: double
- status: bool
+ displayInfo(): void {query}
+ updateDetails(b: string, m: string, y: int, p: double, mil: float): void
+ calculateDepreciation(): double {query}
+ markAsSold(): void
+ getBrand(): string {query}
+ getModel(): string {query}
+ getMileage(): float {query}
+ getPrice(): double {query}
+ getYear(): int {query}

Review

- reviewID: int const
- reviewerID: int const
- sellerID: int const
- ~~int~~ rating: int
- comments: string
- date: string
+ addReview(): void
+ editReview(newRating: string, newComment: string): void
+ deleteReview(): void
+ displayReview(): void

Buyer

- buyerID: int
- name: string
- contact: string
- email: string
- favorites[10]: Car
- ~~favCount: int~~
- market: Marketplace*
+ addFavorite(c: Car): void
+ displayFavorite(): void
+ removeFavorite(index: int): void
+ sendMessage(msg: string): void
+ searchCars(brand: string): void

Listing

- listingID: int
- car: Car
- datePosted: string
- status: bool
- views: int
+ activateListing(): void
+ deactivateListing(): void
+ updateListing(c: Car): void
+ showListing(): void

Message

- messageID: int const
- senderID: int const
- receiverID: int const
- content: string
- timestamp: string
- status: bool
+ sendMessage(): void
+ markAsRead(): void
+ deleteMessage(): void
+ replyMessage(reply: string): void

SearchFilter

- brand: string
- model: string
- minPrice: double
- maxPrice: double
- minYear: int
- maxYear: int
+ matchCar(c: Car const): void {query}
+ applyFilter(cars[]: Car, size: int): void
+ resetFilter(): void
+ showResults(cars[]: Car, size: int): void

Admin

- adminID: int const
- name: string
- email: string
- role: string
- approvedListings[20]: int
- removedListings[20]: int
- approvedCount: int
- removedCount: int
- totalAdmins: int static
+ approveListing(c: Car&): void
+ removeListing(cars[]: Car, size: int&, index: int)
+ displayTotalAdmins(): void static
+ banSeller(s: Seller&): void

MarketPlace

- systemName: string
- allListings[50]: Car
- listingCount: int
- sellers[20]: Seller*
- sellerCount: int
- totalListings: int static
- totalUsers: int static
+ addListing(c: Car): void
+ deleteListing(index: int): void
+ searchListings(): void
+ showAllListings(): void
+ getTotalListings(): int static
+ getListings(): Car*
+ getListingCount: int {query}
+ registerSeller(s: Seller*): void

Seller

- sellerID: int
- name: string
- contact: string
- rating: double
- cars[10]: Car
- carCount: int
- sellerReview: Review
+ addCar(c: Car): void
+ removeCar(index: int): void
+ updateCarListing(index: int, c: Car): void
+ showCars(): void
+ displayInfo(): void {query}
+ showReview(): void

Payment

- paymentID: int
- amount: double
- buyerID: int
- sellerID: int
- method: string
- status: bool
- ~~TAX_RATE: double const~~
+ processPayment(): void
+ refundPayment(): void
+ generateInvoice(): void
+ verifyPayment(): void

Date:
M T W T F S S

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+ removeCar(index: int): void
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+ showReview(): void

Date:

M	T	W	T	F	S	S

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